Gray-Box Modeling for the Optimization of Chemical Processes

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The availability of predictive models for chemical processes is the basic prerequisite for offline process optimization. In cases where a predictive model is missing for a process unit within a larger process flowsheet, measured operating data of the process can be used to set up such models combining physical knowledge and process data. In this contribution, the creation and integration of such gray-box models within the framework of a flowsheet simulator is presented. Results of optimization using different gray-box models are shown for a virtual cumene process.

Keywords: Data selection, Decision support, Flowsheet simulation, Gray-box models, Hybrid models, Optimization, Soft sensing

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1 Introduction

The importance of simulation of entire processes increases in chemical process development. The interaction of experiments (e.g., from a mini-plant) with simulations is beneficial for both. The simulation model helps to improve process understanding especially about the relevant impact factors including their complex relationships and for this reason may support the design of experiments and the evaluation of experimental results. On the other hand, experiments may be used for the validation or if necessary for the adjustment of the simulation model. The goal is to obtain a validated simulation model which might be used for process optimization. Besides a mere cost-driven optimization, the use of multi-criteria optimization, e.g., [1, 2], gains in significance especially with respect to competing sustainability objectives.

One problem for simulations are process units, where rigorous models are lacking or where the effort for the setup of such a model is too expensive. Examples are reactors and decanters. Consider the following procedure for reactors: For a single operating point, the yield can be adjusted so that the simulation model describes this operating point. However, a reliable prediction of other operating points with such models is in general not possible – especially when recycles to the reactor inlet are present. However, if this is done individually for a set of different operating points, one can train statistical models that at least are able to interpolate between the different operating point data. Depending on how much expert knowledge to set up the model is used, the models might even be used for extrapolation purposes. Such gray-box models based on measured operating data are able to eliminate such blind spots. In contrast to a purely empirical, statistical evaluation for the whole process with so-called black-box modeling a lot of previous knowledge is available, which can be used for the setup of gray-box models.

The use of gray-box models (sometimes also called hybrid models) in process simulation is not new. There are applications in the description of bioprocesses, e.g., [3, 4], also with applications of process monitoring and control [5] and also chemical processes like polymer and crystallization processes [6] or for modeling of chemical reactors [7] or used within an ethylene glycol process [8 – 10]. Based on methods of incremental model identification [11, 12] and their successful application for the generation of gray-box models [9, 10], this contribution shows how several tools within a flowsheet simulator have been established to support the different processing steps for gray-box modeling. There have been several tools supporting the modeling (e.g., [6, 8]) but to our knowledge without direct integration into a flowsheet simulator.

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The different processing steps of gray-box modeling are now systematically supported in BASF's in-house simulation software CHEMASIM. In this contribution the workflow of gray-box modeling is described and demonstrated for a cumene process.

The results of the demonstration use case will show that the method is able to achieve improvements of operating conditions for an offline optimization of existing plants. Of course, care has to be taken in the modeling and the validity range. Therefore, methods have been incorporated to reflect the validity range and the prediction error. This helps to restrict the extrapolation step and gives an estimate about the expected uncertainties. The validation of the model predictions with experiments in the plant give the chance to improve the model. Especially a combination with methods for design of experiments and robust optimization can be useful extensions for the modeling-simulation-optimization workflow in the future and can help to achieve faster and more reliable results for improving process performance of existing plants.

2 Gray-Box Modeling for Process Simulation

The approach implemented in BASF's in-house simulation software CHEMASIM is following the concept of incremental model identification (cf. [11, 12]). The use of the approach for hybrid models was already demonstrated by Kahrs and Marquardt [9, 10], but is to our knowledge the first time implemented into a software environment for industrial use to support the workflow of gray-box modeling.

The key idea of gray-box modeling in process simulation is to use available operating data of the process to generate a data-driven model for a process unit with a missing rigorous model. To obtain this data-driven model, a vast number of possible methods from supervised machine learning exist. Quite often however, the inputs and outputs of the process step to be modeled are not measured directly or not easily determined from other measurements. So, before the modeling of the process step can start, these missing data will be estimated based on measured operating data with the help of the already available rigorous model (so-called soft sensing).

Therefore, the first step in gray-box modeling is data selection. The goal is to identify many different representative steady-state operating conditions. Methods to support this data selection step have been described before [13]. At its end, a set of good operating data is available. The application and validity range of the gray-box model will crucially depend on the number of different operating points and the variance in the operating variables (set points).

For the soft-sensing step, a simple surrogate model reflecting all necessary input and output streams of the process step to be modeled is included into the simulation model. For example, in case of generating a data-driven model for an extraction step a component split might be used as a simple surrogate model. In a second step variables of this simple model besides other operating variables of the rigorous part of the simulation model for the entire process (e.g., reflux ratios and heat duties of distillation columns) will be adjusted to each measured operating point to minimize a weighted sum of least squares between measured and simulated properties. For the example with an extraction step, the partition coefficients of the components could be used as adjustment variables besides other operating variables of all other process steps in the simulation model of the entire process.

After this soft-sensing step for the set of operating points all inputs and outputs of the process unit to be modeled are available. This data is at least fulfilling the heat and mass balances. Of course, it is also possible to add additional knowledge. If, for example, for an extraction like mentioned above some partition coefficients are already known, they might be included in the simple model and do not have to be adjusted in the soft-sensing step. After the soft-sensing step all input and output data are available, so it is possible to just focus on this process unit and this data, to analyze it and to set up a data-driven model. Therefore, the big advantage of the incremental modeling is that the model for the process unit can be developed stand-alone.

To support the user in the modeling step different methods have been implemented to analyze the data. This includes correlation analysis (Pearson and Spearman) to identify linear correlations or monotonic relationships between potential model inputs. If so, one of the correlated inputs should be omitted, since otherwise no meaningful estimation of model parameters would be possible. A principal component analysis helps to identify linear combination of the model inputs that explain the data. 2D projections together with a slider concept similar to the one described by Burger et al. [2] to filter data is useful to identify trends and to get ideas about possible model functions and to visualize a comparison of models with data.

A data-driven model ansatz for a set of selected inputs and outputs itself is defined by the user. Of course, the development of a model is still challenging and is hard to be automatized, since the results depend on the selected dependent (outputs) and independent variables (inputs) and the chosen kind of models. Various aspects of the modeling process are described for example in [14]. Therefore, apart from the model itself, emphasis is put on judging its reliability. For this purpose, different options have been implemented: cross-validation, confidence intervals of the estimated parameters and prediction errors. The models are automatically exported in a defined format so that they can be used as user-defined modules in the flowsheet simulator together with the already existing process model.

The validity range of the data-driven model is described based on convex hulls (see e.g. [10]) and on statistical indicators like the prediction error. In our work, inputs are normalized and a function describing the distance of the center of the convex hull reflecting the validity range is generated. This function is exactly equal to zero at the convex hull, greater than zero outside and less than zero inside the convex hull. This function can easily be used directly or as part of a function as a constraint in optimization and ensures that the model will be used only inside or within a user defined percentaged extension of the validity range.

In the following the different steps of gray-box modeling are summarized:

1) Inclusion of a surrogate model for the unknown process step: select a simple surrogate model like a mixer, reactor or component split for use within the simulation model including all necessary input and output streams.
2) Selection of operating data: This can be supported by using CHEMASIM's Plant Data Manager (PDM, [13]). Result of this step is a data base containing data of different operating points together with a mapping of the different measurement tags to CHEMASIM simulation properties.
3) Data pre-processing: Here the previously generated information of the tags and the mapping is used to set up a least-squares optimization problem for model adjustment (cf. [13]). Variables in this optimization are besides operating conditions the variables of the surrogate model like component splits or reaction yields, which are adjusted to each operating point individually in an enumerated (enumeration over operating points) optimization run. This is the soft-sensing step.
4) Modeling: The results of the previous soft-sensing step can be examined within the sensitivity navigator [13] of CHEMASIM using different analysis and visualization tools. Finally, several models correlating different estimated properties (soft sensors) of the unknown process step can be defined. As models either empiric parametric or non-parametric models (like neural networks) can be used.
5) Implementation of gray-box model and optimization: The implementation of the gray-box model is supported. Here the model and functions representing the validity range or estimating the prediction error can be selected. With this extended simulation model, it is possible to perform an optimization of the entire process. Before doing so it is usually recommended to run as a test for the implementation a similar optimization (model adjustment) run as in the pre-processing. To couple the data-driven model with the rigorous model the outputs of the functions of the data-driven model are included in design specifications in the simulation to achieve equality between these model functions and the corresponding properties of the rigorous model, e.g., a split ratio of a splitter should equal the ratio estimated from a gray-box model for partition coefficients. In this run the variables of the surrogate model for the process unit are no longer used as optimization variables. They are now degrees of freedom in the simulation to achieve the design specifications. In this case only operating conditions are used as optimization variables and it can be observed if they differ strongly from the values determined in preprocessing or if principal convergence problems arise.

2.1 Modeling Tools in CHEMASIM

The new established modeling tools are integrated into the sensitivity navigator included in CHEMASIM's results viewer. Here arbitrary properties of the process simulation model (incl. constants, variables, functions, etc. used in the simulation model) can be selected as so-called indicators for the modeling. For these indicators the following list of different tools can be used for analysis:

- Pearson and Spearman correlation analysis: pair-wise analysis of linear (Pearson) or monotone (Spearman) correlation,
- principal component analysis for indicators,
- 2D projections.

The model itself will be created in the "Graybox Model Identification" tab in the sensitivity navigator. It can be used to set up and fit a parametric model or a neural network to the available data, compare different models respectively artificial neural networks and export them for an integration into CHEMASIM as dynamic link library (DLL). First inputs and outputs of the gray-box model have to be selected from the list of indicators. Then a model ansatz can be set up. Predefined linear and quadratic models are available, but it is also possible to set up arbitrary empiric functions. Alternatively, the number of layers and the number of nodes per layer in an artificial neural network can be chosen. Besides the model equations constraints might be formulated for the model adjustment. To fit the model parameters a least-squares problem is solved by using the SQP-LSQ solver from Schittkowski [15].

Model reduction is supported in the tool. Significance levels of the model parameters are determined by eigenvalue/-vector analysis of the Fisher information matrix [16]. If a model seems to be overdetermined, i.e., contains too many parameters compared to the number of equations, the levels give an impression of the significance of parameters and help to decide upon questions like which parameters can be identified and which parameters can be treated as constants.

Furthermore cross-validation is included and confidence intervals of the estimated parameters are given. Model predictions can be visualized in 2D projections.

2.2 Implementation of Gray-Box Model into CHEMASIM Simulation

The selection of the model to be integrated in the CHEMASIM file is supported in the input masks. For the selected models it is possible to integrate the prediction error and the validity range expressed as a normalized convex hull as well. For the chosen model a Fortran file will be written, which will be compiled to a DLL during first use of the gray-box model. Furthermore, in the CHEMASIM input file automatically new external functions for the call of the DLL are included and further variables and functions are defined to reflect the necessary inputs and outputs. All these functions are marked with the model name. The functions of the model outputs should then be used in specifications to be integrated into the simulation. Of course, it is also possible to first integrate and just monitor the outcome of the models for verification and use them in a second step in a specification.

As already mentioned, it is recommended to run a second optimization using the model predictions to set the variables of the surrogate model and only adjust the operating variables. This helps to see if there are convergence problems and it also shows the impact on the operating variables. If they differ in large extent probably the model has to be revisited.

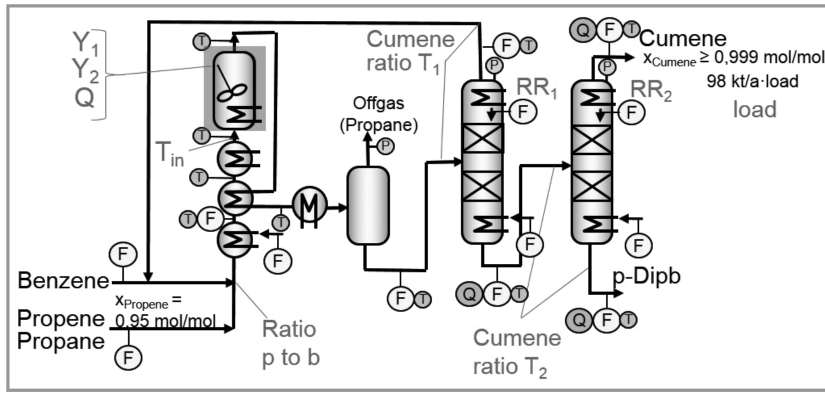

Figure 1. Available measurements and variables of the cumene simulation model for the gray-box modeling of the reactor: In the soft-sensing step for each operating point besides the operating variables (T_{in}, molar mixing ratio propene to benzene, cumene ratio T1 and T2, reflux ratio RR_1 and RR_2, load) the yields Y_1 and Y_2 as well as the heat duty Q are adjusted to minimize the weighted sum of least squares between measured and simulated properties. Circles indicate measurements: F – mass flow, T – temperature, P – pressure, Q – quality, here molar concentration.

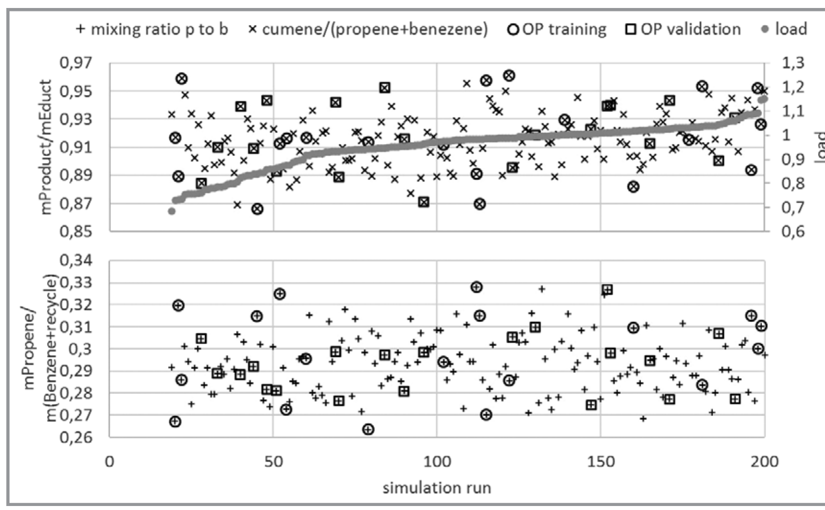

Figure 2. Available and selected artificial operating data of the cumene process (data is based on mass flows including noise). Selection of the data for training was performed to achieve a high variance in the shown properties for the training. Additionally, a set of operating points for validation is selected which also covers a high range of variation.

3 Gray-Box Modeling for a Cumene Process

For an application test an available simulation model for a cumene process [17] has been used as a virtual plant. In contrast to the scheme shown in Fig. 1 the original process model included a tubular reactor consisting of ten segments with known kinetics for the formation of cumene and the side product *p*-di-isopropyl-benzene (DIPB). This original model has been used to generate artificial plant data reflecting different combinations of operating variables covering a load range from approx. 68 to 115 %. How this data was generated is described in Sect. S1 of the Supporting Information. To this artificial data random noise has been added to produce data with standard deviation of 3 % for mass flow (F), 0.01 bar for pressure (P), 1 K for temperature (T) and 0.001 mol mol^{-1} for analytical (Q) measurements. An overview of the available process data range and the selected data points is shown in Fig. 2. 20 operating points have been selected in order to cover a wide operating window for the training. Besides these points, additional 20 points have been selected for the validation of the model, which are also shown in Fig. 2. The operating variables used to produce these two data sets are given in Sect. S2 of the Supporting Information.

The goal of the test was to identify a gray-box process model including a data-driven model for the tubular reactor. Therefore, in the first step as a simple surrogate model a continuous stirred reactor including the two above mentioned reactions was selected. As sketched and described in Fig. 1 besides the operating variables the yields Y_1 and Y_2 of the two reactions and the heat duty Q were adjusted for each selected operating point.

In Tab. 1 and Fig. 3 the average and standard deviations of the soft-sensing step are shown in comparison to the original operating conditions to generate the artificial operating data of the training set. As can be seen the data is identified with good agreement although there was noise added to the artificial data. Especially the large deviations up to 6 % in the yield are not caused by the method but by

Table 1. Deviations of the results of soft-sensing step from the original operating variables of the cumene process for the selected points with and without noise.

Deviations [%]		T_{in}	Mixing ratio p:b	Reflux ratio RR_1	Logarithmic cumene ratio T1	Reflux ratio RR_2	Logarithmic cumene ratio T2	Load
Soft-sensing with noise	Average	0.	0.09	–0.24	0.37	1.73	–1.01	–0.73
	Standard	0.19	1.53	2.36	3.71	2.08	2.66	0.91
Soft-sensing without noise	Average	0	0.14	0.14	–0.04	0.96	–1.13	–0.05
	Standard	0	0.76	0.59	7.17	0.44	1.00	0.25

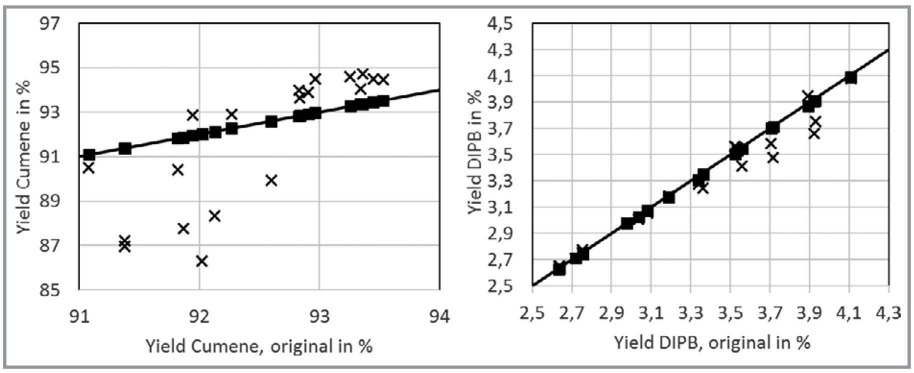

Figure 3. Results for the yields for the formation of cumene and DIPB of the selected operating points for training: results of original model vs results of soft-sensing step with noise (×) and without noise (■).

The results for the yields are used to train two neural networks: one with only one output layer (no hidden layer, cf. Fig. 4, upper part) and one with two layers (one hidden and one output layer, cf. Fig. 4, lower part). As inputs for both networks only the molar flow and the molar concentrations of propene and benzene at the reactor outlet have been used. Outputs which have to be described by the network are the yields for the cumene and the DIPB formation as well as the outlet temperature. Since the inlet temperature was nearly constant in the selected operating points (342 ± 2 °C), it was neglected as input and set to 342 °C.

the measurement noise as has been verified using data without noise (cf. Fig. 3). For the validation data set a similar soft sensing was performed (results are not shown).

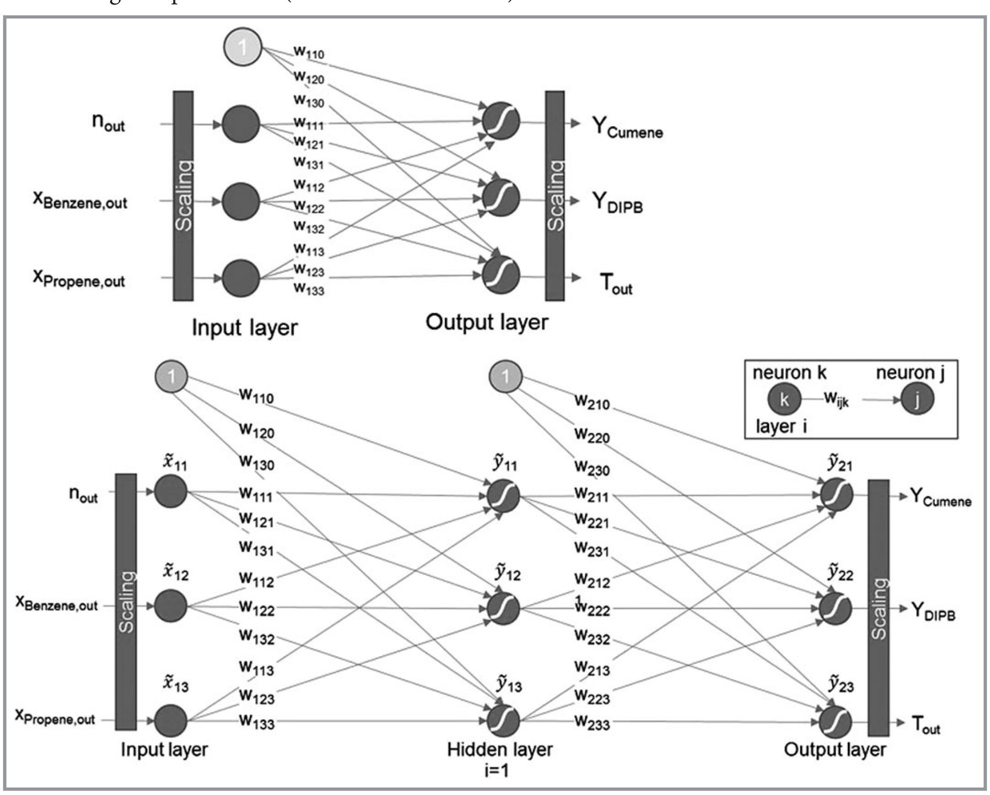

Figure 4. Scheme of the neural network with one layer (upper part) and with two layers (lower part) of the entire process with adjusted operating variables.

The structure of the two neural nets are shown in Fig. 4 (upper part: one layer, lower part: two layers). For the hidden layer of the 2-layer net three neurons were chosen. The inputs x of the training data (see OP training in Fig. 2) in the data range $[x_{\min};x_{\max}]$ are scaled to interval [–1;1].

$$\tilde{x} = \frac{2}{(x_{\max} - x_{\min})}(x - x_{\min}) - 1 \tag{1}$$

For each neuron j of layer i the sum of all weighted (weights w_{ijk}, number $N_{(i-1)}$) inputs $\tilde{x}_{ik}(\tilde{x}_{i0} = 1)$ is used as argument in the sigmoid alpha function (here: $\alpha = 2$ was chosen)

$$\tilde{y}_{ij} = \frac{2}{\left(1 + \exp\left(-\alpha \sum_{k=0}^{N_{(i-1)}} w_{ijk}\tilde{x}_{ik}\right)\right)} - 1 \tag{2}$$

to estimate the outputs. For the output or hidden layers ($i > 1$) the inputs of the neuron j are the outputs of the previous layer

$$\tilde{x}_{ij} = \tilde{y}_{(i-1)j} \tag{3}$$

The outputs of the neural net are in the range [–1;1] and 85 % of this range is scaled to the data range $[y_{\min};y_{\max}]$ used for the training of the net to allow for extrapolation outside this range.

$$y = \frac{1}{0.85}\left(\frac{(y_{\max} - y_{\min})}{2}(\tilde{y} + 1) + y_{\min}\right) \tag{4}$$

Training of the network is supported and based on a randomized initialization of the weights.

In Tab. 2 the coefficient of determination R^2 for the two neural nets are given. The main difference is the R^2 for the yield of DIPB. In Fig. 5 for both neural networks (upper part: one layer, lower part: two layers) the agreement between model predictions with training and validation data is shown. For both models the description of the training and validation data for the yield of cumene and the reactor outlet temperature is in the range of 1 %. For the model with two layers the description of the training data for the yield of DIPB is in the range of 5 % compared to the model with one layer with 10 %. But for both models the validation data is predicted with deviations up to 10 % (with one outlier even above this threshold for the 1-layer model). The 1-layer model has 12 weights (cf. Fig. 4) compared to 24 weights of the 2-layers model which are trained to 20 operating points with three outputs (i.e., 60 output data points). In the following it will be investigated how these models behave when implemented in the simulation model.

Table 2. Coefficients of determination R^2 for the training of the two different neural network models based on the selected operating points (cf. Fig. 2)

R^2	Yield cumene	Yield DIPB	Tout
Neural network 1 layer	0.99	0.92	0.98
Neural network 2 layers	0.99	0.96	0.97

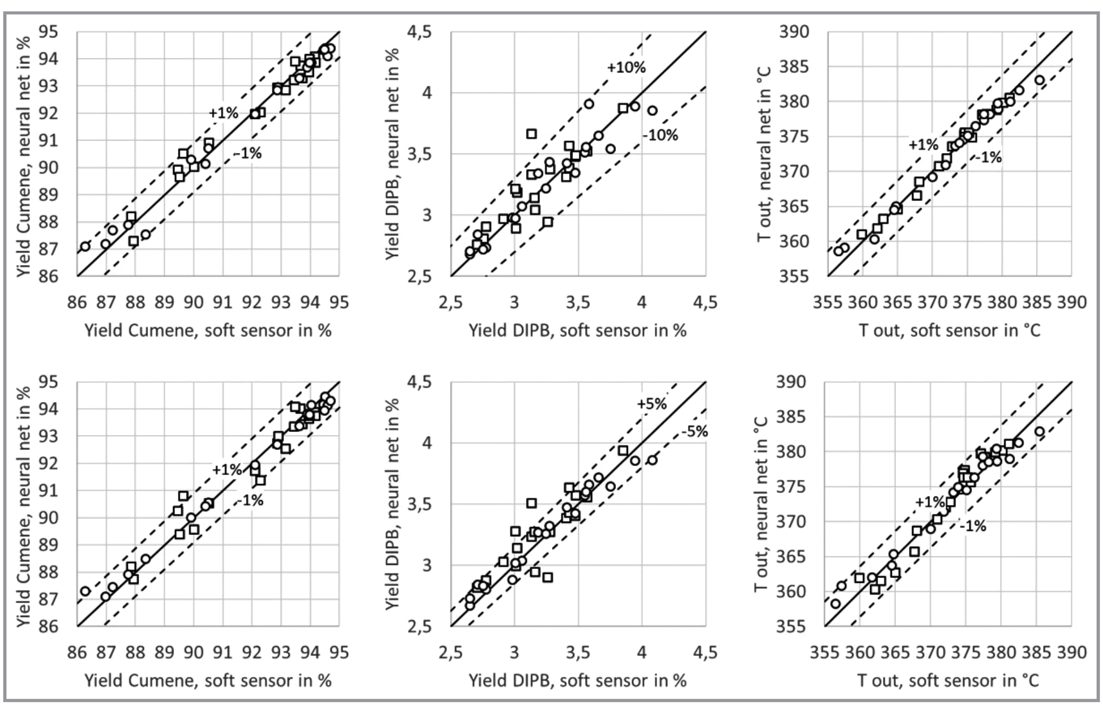

Figure 5. Results for the yields for the formation of cumene and DIPB and for the reactor outlet temperatures of the selected operating points: data from soft-sensing step for the training set (○) and the validation set (□) vs trained neural network with one layer (upper part) and with two layers (lower part).

The different neural networks have been included in the simulation model and these gray-box models were used in a first test to determine the operating variables from the measurements of the selected operating points. The results are shown in Tab. 3 and Fig. 6. As can be seen the models are able to describe the yields within the scatter of the soft-sensing data (cf. Fig. 6). The operating variables are identifiable from the data with nearly the same accuracy as in the previous soft-sensing step (cf. Tab. 3 and Tab. 1, results of soft sensing with noise).

Since the goal of the gray-box modeling is the completion of the simulation model for process optimization, it is interesting to see the results in comparison to the reality or here for this artificial example in comparison to the original model (cf. [17]: case with model VLE2; k1; k2). For this reason, a multi-criteria optimization considering the following objectives has been performed:

- minimize operational expenditure (discounted for 10 years) in relation to the amount of cumene produced (OpEx),
- minimize global warming potential (GWP),
- minimize load, and
- maximize capacity.

The first two objectives are typical sustainability criteria while the last objectives ensure a simultaneous investigation of the minimally and maximally achievable capacity or load.

The variables of the optimization are shown in Fig. 1 and there have been several constraints added to reflect the operating limits of the plant (for details see [17]). In the gray-box models a constant inlet temperature of 342 °C is assumed. In a first application the validity ranges of the neural networks were not included as constraints.

In Fig. 7 the optimization results for both gray-box models (upper part: one layer, lower part: two layers) are shown in comparison to those of the original model. The trend in OpEx vs load is a little bit different for the 1-layer model compared to the original model. However, the maximal deviations in the objectives OpEx and GWP are only about 0.5 %.

To check the quality of the operating conditions (i.e., the optimization variables) resulting from the optimization of these gray-box models, the variables from the gray-box models were used as input for the virtual plant, i.e., the original model, for verification. The results are also shown in Fig. 7 as filled, black circles. As can be seen the deviations for the 1-layer neural network are of the same order of magnitude. So, the prediction of optimal operating conditions for the cumene process based on this simple gray-box model seems to be possible.

The 2-layers model predicts (except for the highest load) the objectives OpEx and GWP in good agreement with the original model, but the verification using the operating conditions of these results shows large deviations in two cases. Furthermore, for the 2-layers model additional constraints to ensure that neither the benzene or the propene concentration becomes negative had to be included.

In a second attempt, the 2-layer model has been used for the same multi-criteria optimization with an additional constraint ensuring that the optimization variables stay in the validity range, which is reflected by a convex hull. The results of this run are shown in Fig. 8. As can be seen the

Table 3. Deviations from the original operating variables of the cumene process for the selected points. Results of the adjustment with two gray-box models including different neural networks (one layer and two layers).

Deviations [%]		T_{in}	Mixing ratio p:b	Reflux ratio RR_1	Logarithmic cumene ratio T1	Reflux ratio RR_2	Logarithmic cumene ratio T2	Load
Neural network 1 layer	Average	–0.15	0.23	–0.01	–0.03	1.69	–1.08	1.21
	Standard	0.24	1.62	2.48	3.94	2.03	2.68	1.01
Neural network 2 layers	Average	–0.15	–0.21	–0.57	1.17	1.67	–1.01	–0.65
	Standard	0.24	1.52	3.12	3.76	1.86	2.72	0.89

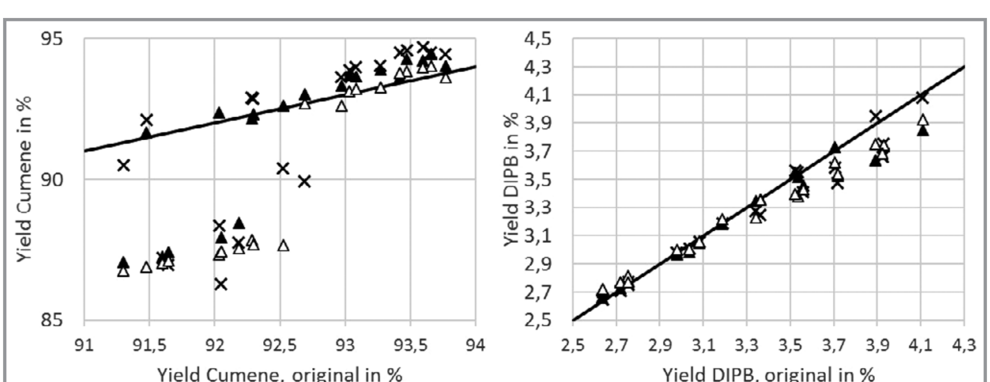

Figure 6. Results for the yields for the formation of cumene and DIPB of the selected operating points for training: results of original model vs results of soft sensing step (×) and gray-box models including neural network with one layer (▲) and with two layers (△) of the entire process with adjusted operating variables.

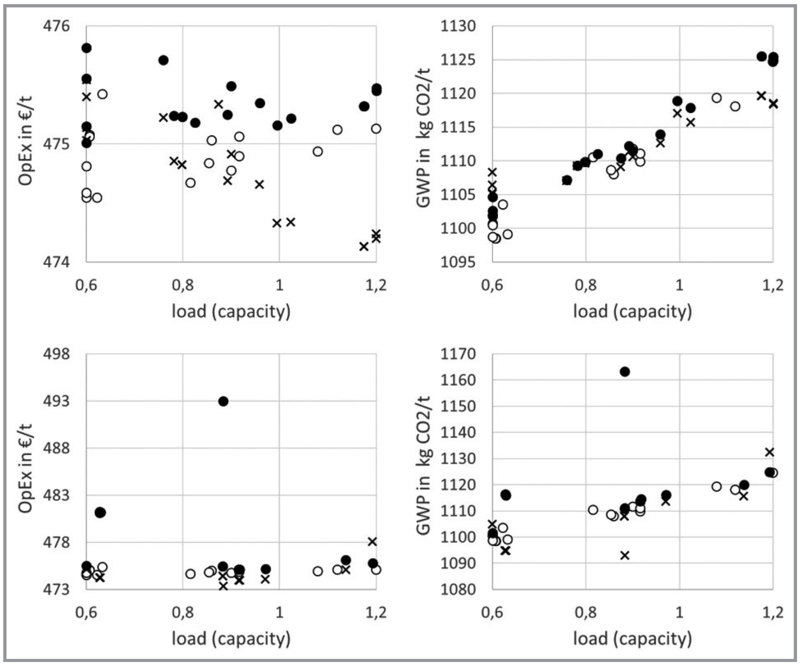

Figure 7. Comparison of the multi-criteria optimization results of the original model (○: cf. [17] Fig. 7, VLE2; k1; k2) and the gray-box model (×) using a neural network. • Verification runs with the original model using operating variables from the optimization with the gray-box model including a neural network. Upper part of the figure shows results of using the neural network with one layer and lower part results with two layers.

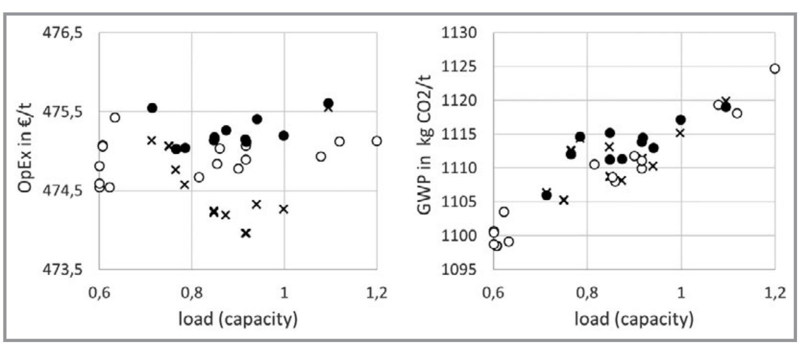

Figure 8. Comparison of the multi-criteria optimization results of the original model (○: cf. [17] Fig. 7, VLE2; k1; k2) and the gray-box model (×) using a neural network with two layers including restriction of the validity range (convex hull). • Verification runs with the original model using operating variables from the optimization with this gray-box model.

agreement between model prediction as well as for the verification of the predicted operating conditions in the virtual plant (i.e., simulation with true parameters) is in close agreement to the results of the optimization with the original model. The results taking the validity range into account do not cover the whole range of the load especially the lowest and highest value of the load reflect points which are on the border of the convex hull.

This is a manifestation of the bias-variance dilemma: Models with less degrees of freedom tend to lead to larger residuals between model outputs and original training data, while showing smaller statistical errors, which is a favorable property for extrapolation purposes. Models with many degrees of freedom are very flexible and interpolate more accurately, while due to their larger statistical error are less adequate for extrapolation. A more formal discussion of this issue can be found, e.g., in [18]. Here, the network with only one layer gives better values also outside the validity range than the neural network with two layers. This was already visible when considering the deviations for the prediction of the validation data. Although the fit of the yield of DIPB was significant better of the two-layer model in comparison to the one-layer model, the prediction of the validation was not. This emphasizes the importance of the validation of models.

4 Conclusions and Outlook

Due to a missing model for certain process steps often process simulations are incomplete and, therefore, cannot be used for optimization of the whole process. The development of gray-box models based on available operating data helps to overcome these limitations. The systematic implementation of tools to support the setup of gray-box models into the workflow facilitates the use of such methods and gives new opportunities for the optimization of entire processes. The example of the cumene process has shown how this could help to identify optimized process conditions. Furthermore, the comparison of two different models demonstrated the importance of the validation of the model prediction with data independent of the training set. A measure for the validity range like the used convex hulls gives the possibility to identify a kind of trust region.

The use of gray-box models in their validity range or with some extrapolation offers the possibility to predict new optimal process conditions. These conditions and their outcome in terms of the optimization objectives have to be proven in plant trials. Such new operation points might be used for the continuous improvement of the gray-box model. Furthermore, the integration of a model-based design of experiments approach would be a good extension in the workflow of this process optimization, since it might help to improve the identifiability of parameters of the gray-box models. The systematic use of the estimated prediction errors as scenarios for a sensitivity analysis or an optimization under uncertainties (either stochastic or robust) gives the chance to investigate the risks connected with the uncertainties of the model. This is especially true when using parametric models. Uncertainty estimation in neural networks is still a challenge [19, 20].

Besides the aspect of developing gray-box models based on plant data this framework is also possible to be used for surrogate modeling (similar for example to [21, 22]). In this context, computational expensive models or sub-models are sampled by a scenario generation method which is also used in sensitivity analysis, e.g., quasi-random sampling using Sobol sequences [13]. The generated simulation data can then be used to develop data-driven surrogate models which then can be implemented and used for optimization as described above. Also, here such models might be iteratively refined (retrained) by using data of verification runs with the original model with the estimated optimization variables. This iteration can be continued until the solution of the optimization between two iteration steps does not change anymore within a small threshold.

Symbols used

N	[–]	number of neurons
R^2	[–]	coefficient of determination
P	[bar]	pressure
Q	[kW]	heat duty
T	°C	temperature
x	[–]	inputs
$\tilde{x}$	[–]	scaled inputs
y	[–]	outputs
$\tilde{y}$	[–]	scaled outputs
Y	[–]	yield
w	[–]	weights
α	[–]	parameter of sigmoid-alpha function

Subscripts

in	at reactor inlet
min	minimal value of data
max	maximal value of data

Abbreviations

b	benzene
DIPB	3-diisopropylbenzene
F	mass flow
GWP	global warming potential
OpEx	operational expenditure
p	propene
P	pressure
Q	quality measurement
RR	reflux ratio
T	temperature

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